COIN BASED MOBILE CHARGING

MINI PROJECT REPORT

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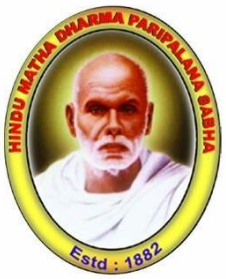
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In

Instrumentation & Control Engineering



Department of Instrumentation & Control Engineering

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DECLARATION

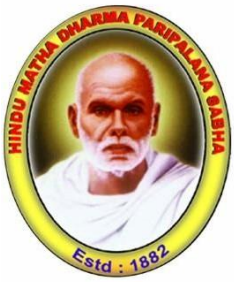
I undersigned hereby declare that the seminar report of **COIN BASED MOBILE CHARGING,** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is bonafide work done by coordinator supervision of Mr. Ajish T. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and /or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom properpermission has not been obtained. This report has not been previously formed for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled '**COIN BASED MOBILE CHARGING'** submitted by **'AZNA V A**' to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements of the award of the Degree of Bachelor of Technology in **INSTRUMENTATION AND CONTROL ENGINEERING** is a bonafide record of the project work carried out by her under my guidance and supervision. This report in any form has not been submitted to any other University orInstitute for any purpose.

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Finally, I thank my Parents without whose blessings I would not have been able to complete the project.

ABSTRACT

Mobile phones are phenomenal in recent years for communication as well as in day-to-day life. Hence, charging the mobile phones has become the greater task. In this paper; we are trying to design a mobile battery charger on coin insertion.

As uses of mobile phones are increasing day by day it need battery life all the time, so in order to use them public charging is needed which would be useful for mobile users. This system will charge the mobile phone for a particular time-period. When the valid coin is recognized, it will start giving power supply to the cell phone throughone of the adapters. We will be using a global charging adapter that would be suitable for all mobile phones. The mentioned system can be implemented in public places likerailway stations, bus stops, hospitals, malls, etc. to avail the services.

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CHAPTER 1
INTRODUCTION

Now a day's mobile phones are become ineluctable part. Power supply is integral part of all electronic systems. Most of the works are done through mobile phones on daily basis, so charging is the basic requirement to operate them. Therefore, the basic idea considered here is to develop a system that will provide charging on coin insertion. The important thing is the above system will be available at public places to be useful at any time. We will be using electricity for power supply as it is all time available source. The user has to insert the coin into coin acceptor and plug the suitable adapter to the mobile phone. The amount of charging will be pre-defined values as mentioned in the microcontroller. This system is easy to install and useful for the long distance travelling peoples.

The coin based mobile charging system charges the mobile phones when the coin is inserted. This system is used by shop owners, rural people and can be implemented in the public places like railway stations, bus stand to provide mobile charging facility. So the coin acceptor recognizes valid coins and then signals the Arduino for further action. If a valid coin is found, it signals the Arduino and then Arduino starts the mobile charging mechanism providing a 5V supply through a power supply section to the mobile phone. The Arduino starts a reverse countdown timer to display the charging time for that mobile phone. Further the user adds another coin, the Arduino adds to the currently remaining time and once again decrements the countdown. This system can be used for smart mobile charging at public places. This coin based mobile charging system will supply the enough amount of charge to the mobile phone and is available on demand in public places.

1.1 Objectives

The objective is by inserting the coin using charge for your mobile phone inpublic places. This is very useful to people who are all using mobile phone without charging condition in public places. In this project, who are all using mobile phones in outside of home are office without charging condition. The coin based mobilephone charger is very useful to that person for using coin to charge for that mobile. A sensor system is used to detect the presence of coin.

CHAPTER 2

LITERATURE SURVEY

Design of Coin Based Mobile Battery Charger

M.S. Vandarajan and et al in 2012 designed a coin based mobile battery charger providing a unique service to the rural public where grid power is not available for partial/full daytime and a source of revenue for site providers.

Coin Based Mobile Charger Using Solar Tracking System

S.B. Sridevi and et al in 2013 designed a coin based mobile charger using solar tracking system in which the mechanical movement of the solar was controlled through the stepper motor.

Ekta Gupta and et al in 2016 designed a coin based solar mobile charger where comparator IC1502 was used which compared the results coming from LDR1 and LDR2.

Intelligent Coin Identification System

Adnan Khashman et al. in 2006 proposed an intelligent coin identification system (ICIS) that uses coin patterns for identification helps preventing confusion between different coins of similar physical dimensions. For recognition of rotated coins of various degrees, ICIS used pattern averaging and neu-ral network. In pre-processing phase ICIS apply thresholding , cropping, compressing, trimming, pattern averaging on images. And then neural network is trained using these images. ICIS used 1 TL and 2 EURO coins in recognition. ICIS used a 3layerback propagation neural network with 400 input neurons, 25hidden neurons and 2- output neuron. The neural network is trained using 20 images out of available 120 coin images. The Accuracy rate achieved was 96.3%.

Indian Coin Recognition System of Image Segmentation

C. M. Velu and P. Vivekanandan et al. in 2009 developed a system for the Indian coin recognition system of image segmentation by heuristic approach and Hough transformation method. This system is developed mainly to classify the coins offered in the Hundi by the devotees of Tirumala Tirupati Devasthanam (TTD), Tirupati, India. The objective is to count money by recognizing the coins and count the total sum based on its value. The system is proposed to design coin recognition by applying heuristic approach, based on the coin table. This table stores parameters of each coin. This project concentrates on affine transformations such as scaling, shearing and soon. This method returns 97of result in recognizing the coin icon. [5] Hussein R. Al-Zoubi et al. in 2010 suggested acoin recognition method using a statistical approach to classify Jordanian coins. There are seven different coins used in Jordan:500fils, 250fils, 100fils, 50fils, 25fils, 10fils, and 5fils. Color and area of a coin was the central feature for sorting. First convert the colored image into grey level and then apply thethreshold value to convert it into black and white image. Then the binary image is cleaned by opening and closing through erosion and dilation, after that calculate the value of each RGBcolor. Then on the basis of these value decisions is made that to which category the coin belongs. Total 1050 experiments,150 for each coin were carried out to examine the proposed system. The Accuracy rate achieved was 9

2.1 Coin Acceptor

The coin acceptor device accepts the coin through coin insertion slot. The sensor inside the coin acceptor device will recognize and validate the coin based on the coin diameter. When the inserted coin is validated, it sends signal to the Arduino for power supply otherwise the coin will be returned back. The sensors in this coin acceptor use the thickness, diameter and fall time of the coins to identify them and it's fully programmable so you're not limited to any particular type of currency. Simply use the buttons and 7-segment display on the side of the unit to select a coin profile, insert a bunch of coin samples (or the same one, over and over) then designate the value. After you've programmed the coin profiles, simply read the serial output of the coin acceptor and it will tell you the value of each coin as they're inserted. It reports these values as binary bytes and the baud rate is selectable on the unit.

This coin acceptor also has a coin reject, so your controller can tell the coin acceptor not to take any more coins, as well as a (very loud) alert beeper. All you need to get the coin acceptor working is a pocket full of change, a 12VDC supply and somewhere for the change to fall after it's accepted.



Fig 2.1: Coin Acceptor

2.2 16*2 LCD Display

LCD is Liquid Crystal Display. Initially it will display ‘please insert a coin’. When the coin is validated, it displays the timing of charging and when coin is not validated it displays a ‘rejected’ message. LCD is attached to Arduino UNO and gets the status of the systems to display on the screen. An electronic device that is used to display data and the message is known as LCD 16×2. As the name suggests, it includes 16 Columns & 2 Rows so it can display 32 characters (16×2=32) in total & every character will be made with 5×8 (40) Pixel Dots. So the total pixels within this LCD can be calculated as 32 x 40 otherwise 1280 pixels.

16 X2 displays mostly depend on multi-segment LEDs. There are different types of displays available in the market with different combinations such as 8×2, 8×1, 16×1, and 10×2, however, the LCD 16×2 is broadly used in devices, DIY circuits, electronic projects due to less cost, programmable friendly & simple to access.



Fig 2.2: LCD Display

2.3 Relay Module

Relay is a switch that open and close circuit electromechanically. Relay control one electrical circuit by opening and closing contacts in another circuit. When the relay circuit detects the undesirable condition with an assigned area and gives the command to the circuit breaker to disconnect the affected area. Thus protects the system from damage. The relay uses an electric current to open or close the contacts of a switch. This is usually done using the help of a coil that attracts the contacts of a switch and pulls them together when activated, and a spring pushes them apart when the coil is not energized. There are two advantages of this system – First, the current required to activate the relay is much smaller than the current that relay contacts are capable of switching, and second, the coil and the contacts are galvanically isolated, meaning there is no electrical connection between them. This means that the relay can be used to switch mains current through an isolated low voltage digital system like a microcontroller.

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Single-Channel Relay Module Applications

- Mains switching
- High current switching
- Isolated power delivery
- Home automation

Specifications

- Digital output controllable
- Compatible with any 5V microcontroller such as Arduino.
- Rated through-current: 10A (NO) 5A (NC)
- Control signal: TTL level
- Max. switching voltage 250VAC/30VDC
- Max. switching current 10A
- Size: 43mm x 17mm x 17mm

The relay uses the current supply for opening or closing switch contacts. Usually, this can be done through a coil to magnetize the switch contacts & drags them jointly once activated. A spring drives them separately once the coil is not strengthened.

By using this system, there are mainly two benefits, the first one is, the required current for activating the relay is less as compared to the current used by relay contacts for switching. The other benefit is, both the contacts & the coil are isolated galvanically, which means there is no electrical connection among them.



Fig 2.3: Relay module

2.4 Arduino UNO

Arduino Uno is a microcontroller board based on ATmega328P (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 Analog inputs, a 16 MHz ceramic resonator (CSTCE16M0V53-R0), a USB connection, a power connector, an ICSP header, and a reboot. button. It contains everything necessary to support the microcontroller, just plug it into a computer with a USB cable or power it up with an AC-to-DC adapter or battery to get started. You can play with your Uno without worrying too much about doing something wrong, in the worst case you can replace the chip for a few dollars and start over. "Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0. The Uno board and version 1.0 of the Arduino software (IDE) were the reference versions of the Arduino, now evolved to newer versions. The Uno board is the first in a series of Arduino USB boards and the reference model for the Arduino platform; For an extensive list of current, outdated, or outdated boards, see the Arduino Board Index.

Specifications

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage (recommended): 7-12V
- In out Voltage (limit): 6-20V
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- PWM Digital I/O Pins: 6
- Analog Input Pins: 6
- DC Current per I/O Pin: 20 mA
- DC current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB (ATmega328P) of which 0.5 KB used by bootloader
- SRAM: 2 KB (ATmega328P)
- EEPROM: 1 KB (ATmega328P)
- Clock Speed: 16 MHz

- LED_BUILTIN: 13
- Length: 68.6 mm
- Width: 58.4 mm
- Weight: 25 g

The word "uno" means "one" in Italian and was chosen to mark the initial release of Arduino Software.The Uno board is the first in a series of USB-based Arduino boards; it and version 1.0 of the Arduino IDE were the reference versions of Arduino, which have now evolved to newer releases. The ATmega328 on the board comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer.While the Uno communicates using the original STK500 protocol, it differs from all preceding boards in that it does not use the FTDI USB to serial driver chip.

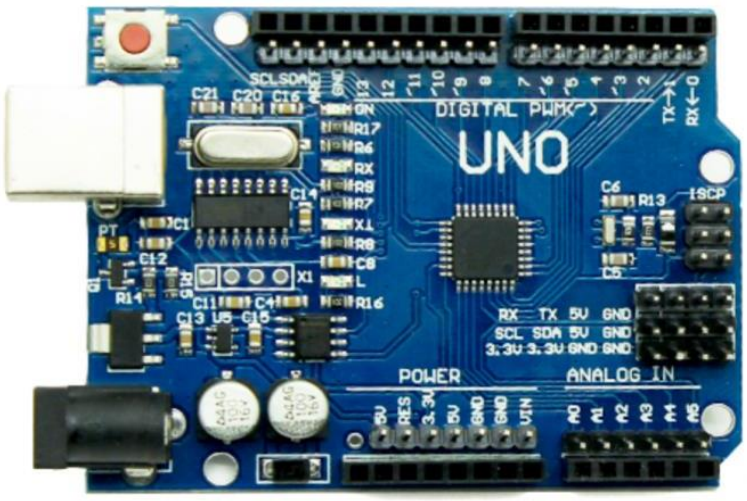


Fig 2.4: Arduino UNO

2.5 SMPS

It is an efficient and effective source of power. It is a device in which energy conversion and regulation is provided by power semiconductors that are continuously switching “on” and “off” with high frequency. The switching regulator does the regulation in the SMPS. It turns the current supply to a smoothing capacitor on and off.SMPS dissipates very less energy as heat. It works efficiently and is only low heat. Its output of constant voltage ensures the stability of the power supply. The switching power supply is provided with protection against over-voltage, short circuit and overload. The anti-interference function guaranteed a long lifetime for the LED lamps.Switching power supply products are widely used in: industrial automation control, military industry equipment, scientific research equipment, security, monitoring, entrance guard, building intercom, LED lighting, advertising lights, industrial control equipment etc.

Specifications

- Input Voltage: AC 100 - 264V 50 / 60Hz
- Output Voltage: 5V DC, 5A
- Protections: Overload / Over Voltage / Short Circuit
- Auto-Recovery After Protection
- Universal AC input / Full range
- 100% Full Load Burn-in Test
- Cooling by Free Air Convection
- High Quality and High Performance
- LED power supply with a metal body for hidden installation for LED lighting
- Design with Built-in EMI Filter, improve signal precision.
- Certifications: CE & RoHs
- No Minimum Load.
- Compact Size Light Weight.
- High Efficiency, Reliabiltiy & low energy consumption

- Category - Switch Mode Power Adaptor (SMPS)
- MTBF >50,000 hours
- Output Type - DC
- Output - 5Volts 5Amp
- Shell Material: Metal Case / Aluminum Base
- Color: Sliver
- In-door use only.



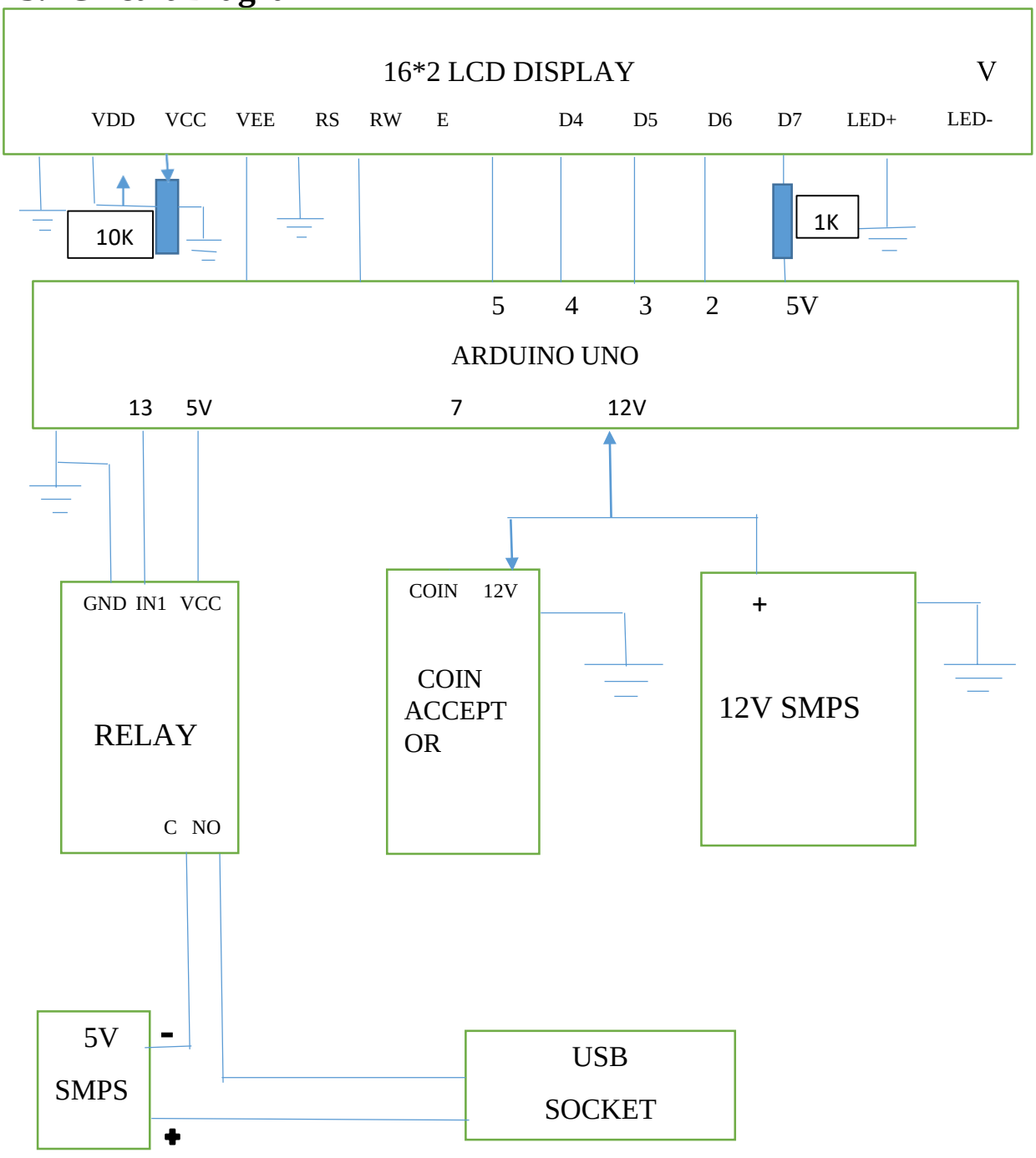
Fig 2.5: SMPS

CHAPTER 3

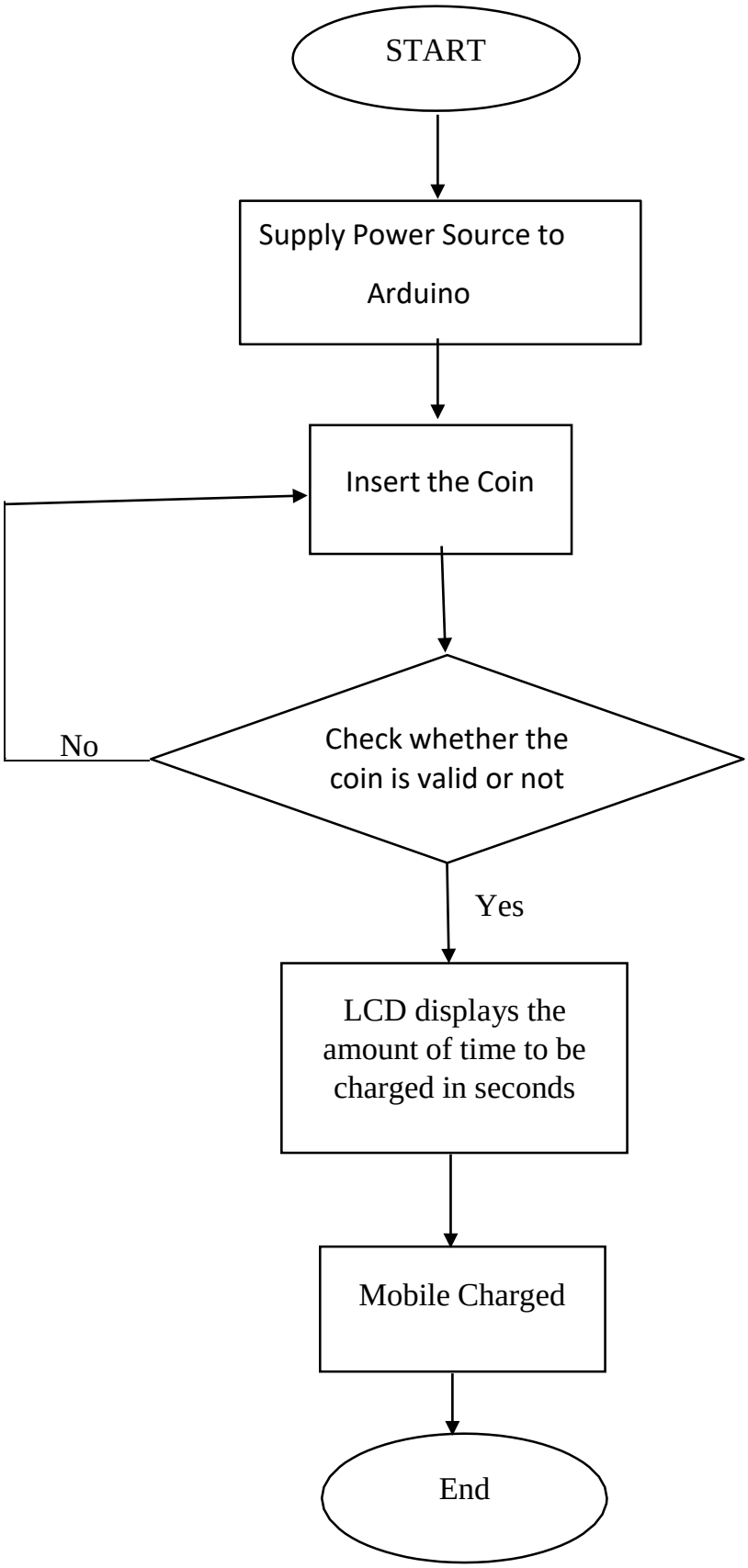
PROPOSED SYSTEM

This is the smart coin based mobile charging system that charges your mobile for particular amount of time on inserting a coin. The system consists of a coin recognition module that recognizes valid coins and then signals the microcontroller for further action. If a valid coin is found it signals the microcontroller and then starts the mobile charging.

3.1 Circuit Diagram



3.2 Program Flow Chart



The user inserts a coin to the coin insertion slot. The sensor is attached to the coin insertion slot and the coin is validated based on the diameter of the coin inserted. Initially the LCD display a message as “Please insert coin”. If the inserted coin is valid,the message is displayed in the LCD and signal is sent to the Arduino. If the coin is notvalid, it is returned back. When the coin is accepted, the Arduino and relay is activatedand the battery starts getting charged by the software of relay. The system performs according to the input signal from the circuit. Based on the diameter of the coin, the coin is either accepted or rejected. If the coin is accepted, it sends signal to Arduino along with LCD interface. Once the Arduino receives the signal from the coin insertion slot, it sends signal to the relay. The relay generates the voltage of 5v, which in turn charges the mobile phone through the mobile USB terminal.

The LCD connected displays the messages as and when required. Initially, when the mobile charger is connected the LCD displays as, “Please insert coin”. When the mobile phone is charging, it displays “Charging” and the duration of charging based on the coin inserted. This coin based mobile charger draws power from the Arduino through relay. The voltage is regulated based on the type of the mobile phone connected for charging.

3.4 Advantages

- Simple and hand efficient
- Less expensive
- Reduced man power
- Simple to operate
- Low power consumption
- Installation is easy

3.5 Disadvantages

- It requires a power supply for operation
- Careful for the instrumental error

3.6 Applications

- It can be used for different types of mobiles
- Used for emergency charging purpose
- It can be installed in railway stations, bus stops and rural areas
- It can be installed in office and colleges for pay charging facility
- Useful to public for using coin to charge the mobile phone at any place

3.7 Future scope

The project can be used in the following areas: -

- Railway station: This type of project used in railway stations
- Shop: Coin based mobile charging is install any shop and earn money
- Rural areas: Installing in rural areas where the power grid is not available at any time
- Public place: This project is very useful to people who are using mobile phone without charging condition in public places

CHAPTER 4
RESULTS AND DISCUSSION

In this system, we have implemented the simple and hand efficient mobile charger which helps the user, charge their phones during urgent needs. This system is very helpful to the users who are all using mobile phone without charging conditions in public places. This system simple to use and is less expensive. As uses of mobile phones are increasing day by day it need battery life all the time, so in order to use them public charging is needed which would be useful for mobile users. This system will charge the mobile phone for a particular time-period. When the valid coin is recognized, it will start giving power supply to the cell phone through one of the adapters. We will be using a global charging adapter that would be suitable for all mobile phones. The mentioned system can be implemented in public places like railway stations, bus stops, hospitals, malls, etc. to avail the services.

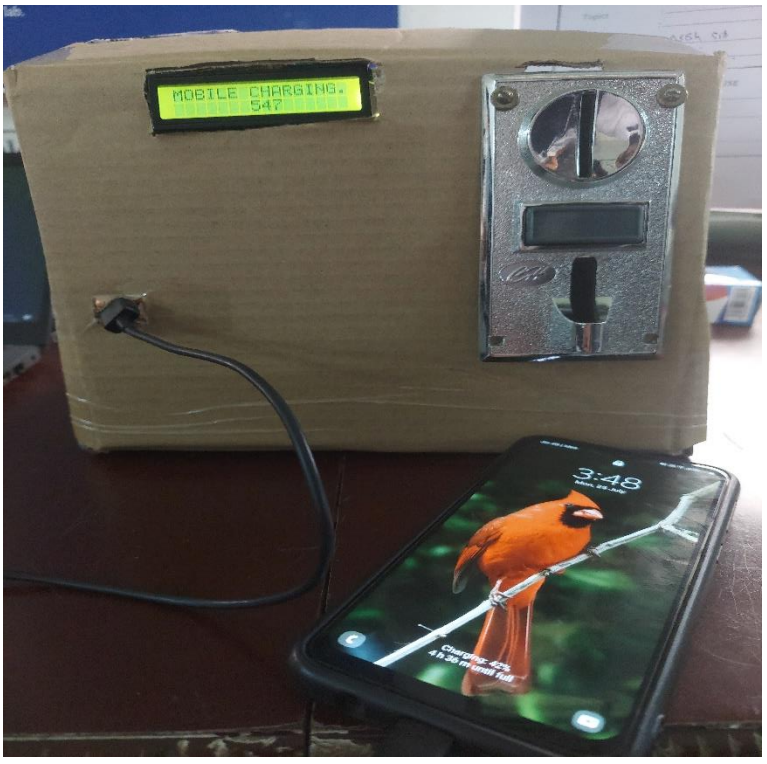


Fig 4.1 Project Demo

CHAPTER 5

CONCLUSION

This project describes the new way of providing charging services to the public. It would be of less cost because conventional grid power is used and beneficial to the long distance travelers. This coin based mobile charging system can be installed at various public places for the convenience of mobile users. The objective of this paper is to help mobile users by providing coin based charging that would be easily available whenever they need it.

The Arduino based smartphone charging controller is implemented here, which is a simple project to control the amount of time your mobile phone charges. With this project, you can connect your phone to the charger, set the time for which your phone charges, and forget about it, as the project will automatically disconnect the power from the charger. This project is very useful for people who tend to charge the phone overnight or for those who often forget that they plugged the phone into the charger.

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APPENDIX

SOFTWARE USED: THE ARDUINO IDE

The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. It is used to write and upload programs to Arduino compatible boards, but also, with the help of third party cores,other vendor development boards.The source code for the IDE is released under the GNU General Public License, version

The Arduino IDE supports the languages C and C++ using special rules of code

structuring.The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub main() into an executable cyclic executive program with the GNU tool chain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware. By default, avrdude is used as the uploading tool to flash the user code onto official Arduino boards.

SOURCE CODE

```
#include <LiquidCrystal.h>


// initialize the library with the numbers of the interface pins

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

int i;

void setup() {
```

```
  lcd.begin(16, 2);
  pinMode(13, OUTPUT);
  pinMode(8, OUTPUT);
  pinMode(7, INPUT_PULLUP);
  lcd.print("SOLAR AND COIN ");
  lcd.setCursor(0, 1);
  lcd.print("BASED MOBILE");
  delay(1500);
  lcd.setCursor(0, 0);
  lcd.print("              ");
  lcd.setCursor(0, 1);
  lcd.print("     CHARGER      ");
  delay(1000);
  lcd.setCursor(0, 0);
  lcd.print("INSERT THE COIN");
  lcd.setCursor(0, 1);
  lcd.print("                ");

}

void loop() {
  lcd.setCursor(0, 0);
  lcd.print("  INSERT  COIN  ");
```

```
 lcd.setCursor(0, 1);
 lcd.print("                ");
 if(digitalRead(7)==HIGH)
 {
    lcd.print(" COIN INSERTED ");
    delay(1000);
    lcd.setCursor(0, 0);
    lcd.print("MOBILE CHARGING.");
    digitalWrite(8,LOW);
    for(i=599;i>=0;i--)
    {
     lcd.setCursor(0, 1);
     if(i>9)lcd.print("               ");
     else lcd.print("                ");
     lcd.print(i);
     delay(1000);
     digitalWrite(13,LOW);
    }
 }
 digitalWrite(8,HIGH);
 digitalWrite(13,HIGH);
}
```